Kinds of collisions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate momentum conservation from kinetic-energy conservation.

SUCCESS CRITERIA - I can...

1. Identify the system and external-impulse assumption.
2. Recognize sticking as perfectly inelastic.
3. Use kinetic energy to distinguish elastic from inelastic.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A positive triangular force pulse lasts 0.3 s and peaks at 50 N. Find impulse and velocity change for a 0.6 kg object.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$; $K = \frac{1}{2}mv^2$.
Negligible external impulse. Elastic collisions also have $K_{total,\,i} = K_{total,\,f}$; sticking transforms some kinetic energy.

Worked reasoning: A 1 kg cart at 4 m/s joins a 1 kg cart at rest. Initial momentum is 4 kg·m/s. Together, 2 kg move at 2 m/s; sticking is one possible collision outcome.

Your turn: Draw before and after snapshots with masses and velocities.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch before/after for a collision where two carts stick. Label one common final velocity, not two different final velocities.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart traveling at +3 m/s sticks to a stationary 1 kg cart on a low-friction track. Draw and label the collision. Find the common final velocity, kinetic energy before and after, and fraction of initial kinetic energy transformed.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A classmate says energy is destroyed in an inelastic collision. Rewrite the claim accurately.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A projectile launches horizontally at 6 m/s from height 45 m. Use g=10 and no drag. Find flight time, range and final vertical velocity.
Label separate horizontal and vertical motion.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two equal masses collide elastically head-on, one initially at rest. Give their final velocities relative to initial moving speed u. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. For the same initial state, test final velocities (0, 4), (1, 3) and (2, 2) m/s for momentum conservation. Which pairs represent sticking? Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3